ReHQJ: the HexOpt Paper and conspire

autotwin/automesh issue #768

October 14, 2026

Sources

This note compares the per-hex objective of HexOpt in two sources.

- **Paper.** H. Tong and Y. J. Zhang, "Fast and Robust Hexahedral Mesh Optimization via Augmented Lagrangian, L-BFGS, and Line Search," arXiv:2410.11656, Equations (2.5) and (2.6).
- **conspire.** Branch `msj` at commit `515c78cee`, draft mrbuche/conspire.rs#251, file `src/geometry/mesh/quality/improve/msj/mod.rs`, function `terms`.

h is one hex element of the mesh H, and $\bar{e}$ is its average edge length. $SJ(h)$ is the minimum of the scaled Jacobian over nine points of h: its eight corners and its body center (paper, Section 2.1). $J(h)$ is the minimum of the Jacobian over the same nine points. conspire computes both minimums in `terms`, each over all nine points. Θ is the threshold. Both sources hold $\bar{e}$ fixed when they take the gradient.

Definitions

The paper defines

$$ReHQJ_{\text{paper}}(h,\Theta) = \begin{cases} J(h)/\bar{e}, & J(h) \le 0, \\ SJ(h)\,\bar{e}^2, & J(h) > 0,\ SJ(h) \le \Theta, \\ \Theta, & SJ(h) > \Theta. \end{cases}$$

conspire implements

$$ReHQJ_{\text{conspire}}(h,\Theta) = \begin{cases} J(h)/\bar{e}, & J(h) \le 0, \\ SJ(h)\,\bar{e}^2, & J(h) > 0,\ SJ(h) \le \Theta, \\ \Theta\,\bar{e}^2, & SJ(h) > \Theta. \end{cases}$$

The two differ only in the third case.

Units

$J = \det(e_0, e_1, e_2)$, where each edge vector has units of length. SJ and Θ have no units.

case	paper	conspire
$J \le 0$	length^2	length^2
$J > 0$, $SJ \le \Theta$	length^2	length^2
$SJ > \Theta$	**none**	length^2

Behavior at the threshold

1. **Continuity.** At $SJ = \Theta$, the second case gives $\Theta\,\bar{e}^2$. The conspire form gives the same value, so the objective is continuous at Θ. The paper's form gives Θ, so the objective jumps by $\Theta(\bar{e}^2 - 1)$ unless $\bar{e} = 1$. Neither form is differentiable at Θ. The slope drops to zero there in both, by design.

2. **The sign of the jump.** The minimized function is

$$L = -\sum_h ReHQJ(h, \Theta) + \cdots,$$

   as in Equation (2.8). Under the paper's form, a hex that rises above Θ changes L by $+\Theta(\bar{e}^2 - 1)$. For $\bar{e} > 1$, L rises. The Armijo condition requires L to fall, so the jump counts against every step that lifts such a hex above Θ. For $\bar{e} < 1$, L falls, so the jump rewards the crossing instead. The sign depends on the length unit of the mesh.

   - On the snapped bone octree in the reference frame, $\bar{e}$ runs from 0.644 to 2.81, with a median of 1.253. The same mesh has hexes on both sides of $\bar{e} = 1$.
   - On the scale-5 bone mesh in the STL's own units, $\bar{e}$ runs from 0.009 to 0.034.

   Under the conspire form, the crossing changes L by zero.

3. **Units.** Under the conspire form, all three cases have units of length^2. conspire's types enforce this: each term is a `Quantity<Area>`.

For a hex that stays above Θ through a step, both forms are constants. They give the same gradient and the same energy differences. The forms differ only when a hex crosses Θ.